Primitive Denominator Cancellation Spectra and Local Densities
for Unimodular Word Forms and Quadratic Swap Returns

Alp Eren Bütün
Independent Researcher, Türkiye
ORCID: 0009-0001-2479-6037

25 October 2026

Abstract

For a primitive integral vector u and a unimodular integer matrix M, we reconstruct the unique symmetric positive-definite determinant-one rational form $P(M,u)$ sending u to M^Tu. Its least integral denominator is $f/\gcd(f, M_{12}-M_{21})$, where $f = u^TM^Tu > 0$. For each nonsymmetric nonnegative M we classify every cancellation factor attainable on the strictly positive primitive cone and prove infinitely many realizations. More precisely, only primes 1 (mod 4) and at most one factor of 2 cancel, subject to an exact even-diagonal exception. We further determine the *exact rational natural density* of each factor: a universal Euler product depending only on $|M_{12}-M_{21}|$ and the exceptional parity, with an $O_M(X\log X)$ error for vectors in $[1,X]^2$. We retain the integrality–palindrome criterion, the classical Fricke commutator-trace condition and a restricted word-orbit swap-return construction. The arithmetic spectrum and density are proved by local quadratic-form classification, Chinese remaindering and primitive lattice-point counting; historical novelty is not asserted.

1 Introduction

Matrix words, primitive lattice vectors, positive binary quadratic forms, and palindromes occur throughout the classical Cohn–Markoff and Christoffel theories; see Cohn [1] and Reutenauer [5]. Our question is more specific: given an integral unimodular word state M^Tu, can one recover a canonical determinant-one symmetric quadratic observable, determine *exactly* when it is integral, and prescribe the unique word at which two adjacent orders have equal quadratic value?

The main new research direction of this revision is an exact finite cancellation spectrum and a universal Euler-product frequency law in the positive primitive cone. Both build on elementary local quadratic-form and lattice counting, whose general theory is classical. We also retain three matrix-word statements that are logically distinct: (i) an integral denominator identity for an arbitrary $SL_2(\mathbb{Z})$ matrix; (ii) a common-factor condition on a whole family of swap quadratic forms; and (iii) a zero-set statement on an injectively decoded *word orbit*. The last does not follow from a discriminant-square test alone. The proofs are elementary and self-contained. The commutator condition belongs to classical Markoff theory, and we distinguish it explicitly from the canonical exact-denominator formula and from the orbit-level return statements. Existing Markoff quadratic-form theory is extensive, including [2, 3]; no claim of historical firstness of the remaining statements is made.

Write $J = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, so $\det(a,b) = -a^TJb$. For rational symmetric P, put $Q_P(v) = v^TPv$. For a rational matrix P, let $\mathrm{den}(P)$ denote the smallest positive integer D such that DP has integral entries.

Contributions and novelty

The paper develops five connected results with distinct scopes. First, it gives a canonical, unique determinant-one symmetric rational reconstruction and an exact least-denominator formula for any primitive seed with positive pairing. Second, for nonsymmetric nonnegative unimodular matrices it determines every attainable cancellation factor on the strictly positive primitive cone, including the obstruction to primes 3 (mod 4), the bound on the 2-adic factor, and the even-diagonal exception. Third, it computes the full local-density distribution of those factors and an Euler-product natural-density law with an effective lattice-point error term. Fourth, it connects the classical parabolic Fricke commutator condition to a universal common-factor property of swap quadratic observables. Fifth, with positivity and injective word-state decoding, it constructs quadratic observables having one prescribed swap-zero word and retains the integrality–palindrome criterion.

The local splitting arguments, primitive lattice counting, Fricke trace identity, and classical Markoff and palindrome background are not claimed as new techniques or discoveries. The paper's contribution is the stated combination of exact denominator, spectrum, density, common-factor and return statements in their respective models. No historical priority claim beyond the explicitly proved assertions is inferred from a limited literature comparison.

2 Canonical reconstruction and its exact denominator

Theorem 2.1 (Unique canonical quadratic reconstruction)**.** *Let* $M = \left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right) \in SL_2(\mathbb{Z})$ *and let* $u \in \mathbb{Z}^2$ *be primitive with*

$$q = M^T u, \qquad f = u^T q > 0, \qquad z = Ju.$$

Then

$$P(M,u) = \frac{qq^T + zz^T}{f} \tag{1}$$

is the unique symmetric rational matrix of determinant 1 *satisfying* $Pu = q$*. It is positive definite and its least integral denominator is*

$$\boxed{\operatorname{den}(P(M,u)) = \frac{f}{\gcd(f, B-C)}.} \tag{2}$$

Proof. The vectors q and z are independent because $\det[q,z] = q^T u = f > 0$. Equation (1) is therefore positive definite and has determinant $\det[q,z]^2/f^2 = 1$. Since $z^T u = 0$ and $q^T u = f$, we have $Pu = q$. Any other symmetric solution is of the form $P + tzz^T$: the difference must annihilate u, so its image is in $\mathbb{Q}z$. The rank-one determinant lemma then yields

$$\det(P + tzz^T) = 1 + t\, z^T \operatorname{adj}(P) z.$$

The last scalar is strictly positive, forcing $t = 0$ for determinant 1.

For the denominator calculation, extend the primitive u to an integral unimodular basis $T = [u,t] \in SL_2(\mathbb{Z})$. Set

$$T^T M^T T = \begin{pmatrix} f & h \\ g & k \end{pmatrix}, \qquad T^T P T = \begin{pmatrix} f & g \\ g & \frac{g^2+1}{f} \end{pmatrix}.$$

The second matrix has the stated first column and determinant one. Integral unimodular congruence preserves the least common denominator. Consequently

$$\operatorname{den}(P) = \frac{f}{\gcd(f, g^2+1)}.$$

From $fk - gh = 1$ we obtain $g^2 + 1 \equiv g(g-h) \pmod{f}$. Also $\gcd(f,g) = 1$, so $\gcd(f, g^2+1) = \gcd(f, g-h)$. Finally,

$$g - h = u^T(M - M^T)t = (B - C)\det(u,t) = B - C.$$

Substitution completes the proof. □

Corollary 2.2 (Integrality is symmetry in the positive cone)**.** *If $M \in SL_2(\mathbb{Z}_{\geq 0})$ and $u = (r,s)^T$ is primitive with $r, s > 0$, then*

$$P(M,u) \in \mathrm{Sym}_2(\mathbb{Z}) \quad \Longleftrightarrow \quad B = C.$$

In the integral case $P(M,u) = M$.

Proof. We have $f = Ar^2 + (B+C)rs + Ds^2 \geq A + B + C + D > |B - C|$, since $A, D \geq 0$ and $AD - BC = 1$ imply $A + D > 0$. Hence $f \mid (B - C)$ exactly when $B = C$. Theorem 2.1 yields integrality. If M is symmetric, it is positive definite (its diagonal entries are positive and its determinant is one), and it satisfies $Mu = M^T u = q$, so uniqueness gives $P = M$. □

Remark 2.3 (Sharpness of the seed assumption)**.** *The strict positivity of both seed coordinates cannot be discarded. With $u = (1,0)^T$ and $M = \left(\begin{smallmatrix} \frac{1}{2} & \frac{1}{3} \end{smallmatrix}\right)$ one obtains the integral form $P = \left(\begin{smallmatrix} \frac{1}{1} & \frac{1}{2} \end{smallmatrix}\right)$ although $B \neq C$.*

Corollary 2.4 (Palindrome detection)**.** *Let $G, H \in SL_2(\mathbb{Z}_{\geq 0})$ be symmetric, and suppose the matrix word map $w \mapsto M(w)$ is injective. For a positive primitive u, put $P_w = P(M(w), u)$. Then*

$$P_w \text{ has integral entries} \quad \Longleftrightarrow \quad w = w^R.$$

Proof. The generators are symmetric, hence $M(w)^T = M(w^R)$. By Corollary 2.2, integrality is equivalent to $M(w) = M(w)^T = M(w^R)$, and injectivity finishes the proof. □

3 The complete primitive cancellation spectrum

For the remainder of this section let

$$M = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \in SL_2(\mathbb{Z}_{\geq 0}), \qquad B \neq C, \qquad k = |B - C| > 0,$$

$$F_M(X,Y) = AX^2 + (B+C)XY + DY^2.$$

For every strictly positive primitive column $u = (x,y)^T$, put $c_M(u) = \gcd(F_M(x,y), k)$; Theorem 2.1 gives $\operatorname{den} P(M,u) = F_M(x,y)/c_M(u)$. Write v_ℓ for the usual ℓ-adic valuation.

Theorem 3.1 (Attainable cancellation factors)**.** *The exact set of attained cancellation factors is*

$$\{c : c \mid k, \quad v_2(c) \leq 1, \quad \ell \mid c,\ \ell \text{ odd prime} \Rightarrow \ell \equiv 1 \pmod 4, \quad (A, D \text{ both even} \Rightarrow 2 \mid c)\}. \tag{3}$$

Each admissible c is attained by infinitely many strictly positive primitive vectors u. The exceptional even-diagonal case forces $v_2(k) = 1$.

Proof. Choose an integral unimodular basis $T = [u,t]$ and use the proof of Theorem 2.1: in its notation,

$$\gcd(F_M(u), k) = \gcd(f, g^2 + 1).$$

An odd prime divisor of the right-hand side must admit a square root of -1, hence is $1 \pmod 4$. Also $v_2(g^2+1) \leq 1$: if g is even then g^2+1 is odd, while if g is odd then $g^2+1 \equiv 2 \pmod 8$. Thus

no cancellation factor can contain 4. If k is even, then $B+C$ is even and $F_M(x,y) \equiv Ax+Dy \pmod 2$. When A, D are both even, F_M is always even. Since $AD-BC=1$, in this case B, C are odd and $4 \nmid B-C$: otherwise $BC \equiv 1 \pmod 4$ contradicts $AD-BC=1$. This proves necessity.

For sufficiency fix $c \mid k$ as in (3). At an odd prime power $\ell^e \| k$, the quadratic discriminant

$$\Delta = (B+C)^2 - 4AD = (B-C)^2 - 4 \equiv -4 \pmod{\ell^e}$$

is a unit. If $\ell \equiv 3 \pmod 4$, the form is anisotropic on nonzero vectors modulo ℓ, and its value is a unit on every primitive class. If $\ell \equiv 1 \pmod 4$, its nondegenerate split reduction can be changed by an invertible linear substitution over $\mathbb{Z}/\ell^e$ to a unit times XY. Every value of $v_\ell(F_M)$ between 0 and e (truncated at e) therefore occurs on a primitive class. At 2, either A, D are both even, when $v_2(k)=1$ and every primitive class gives $v_2(c)=1$; or the nonzero linear form $Ax+Dy \pmod 2$ has primitive classes of both parities. In the latter case no primitive class can yield $v_2(F_M) \geq 2$ when $4 \mid k$, by the necessary condition just established. Thus every specified admissible local valuation is attained. CRT supplies a primitive residue class $(a,b) \pmod k$ with $\gcd(F_M(a,b),k)=c$.

Finally every class $(a,b) \pmod k$ with $\gcd(a,b,k)=1$ contains infinitely many positive primitive integer vectors. Fix any positive $x \equiv a \pmod k$. For primes dividing x and k, the prescribed $b \pmod k$ is nonzero; for the remaining finitely many primes dividing x, choose $y \pmod \ell$ to be nonzero, in addition to $y \equiv b \pmod k$. CRT gives infinitely many such positive y, all coprime to x. The cancellation factor is unchanged modulo k. □

4 Exact local densities and coefficient universality

Let

$$\mathcal{V}_X = \{(x,y) : 1 \leq x,y \leq X,\ \gcd(x,y)=1\}, \qquad N_c(X) = \#\{u \in \mathcal{V}_X : c_M(u)=c\}.$$

For each $\ell^e \| k$, let $\pi_{\ell,e}(t)$ be the following probability distribution on the truncated valuation $t = v_\ell(c)$ for a uniformly chosen primitive pair modulo ℓ^e.

local case	$\pi_{\ell,e}(t)$
$\ell \equiv 3 \pmod 4$	$\pi(0)=1$
$\ell \equiv 1 \pmod 4$	$\pi(0)=\dfrac{\ell-1}{\ell+1}, \quad \pi(t)=\dfrac{2(\ell-1)}{(\ell+1)\ell^t}\ (1 \leq t < e), \quad \pi(e)=\dfrac{2}{(\ell+1)\ell^{e-1}}$
$\ell=2,\ A,D$ both even	$e=1, \quad \pi(1)=1$
$\ell=2$, otherwise	$\pi(0)=\dfrac{2}{3}, \quad \pi(1)=\dfrac{1}{3}.$

All unlisted values have probability zero. For $c \mid k$, define

$$\rho_{M,k}(c) = \prod_{\ell^e \| k} \pi_{\ell,e}(v_\ell(c)), \tag{4}$$

with value zero when $c \nmid k$. For $k=1$, the empty product is 1.

Theorem 4.1 (Exact cancellation densities)**.** *For each fixed nonsymmetric $M \in SL_2(\mathbb{Z}_{\geq 0})$ and each positive integer c,*

$$\boxed{N_c(X) = \frac{6}{\pi^2}\rho_{M,k}(c)X^2 + O_{M,k}(X\log(2X)).} \tag{5}$$

In particular the relative frequency among positive primitive vectors has limit $\rho_{M,k}(c)$, and this is positive precisely for the factors in Theorem 3.1. Apart from the exceptional case where both diagonal entries are even, the entire distribution depends only on $k=|B-C|$, not on any other entry of M.

Proof. Fix $\ell^e \| k$. The nonsingular discriminant -4 modulo ℓ makes F_M anisotropic when $\ell \equiv 3 \pmod 4$. For $\ell \equiv 1 \pmod 4$, an invertible transformation over $\mathbb{Z}/\ell^e$ takes F_M to a unit times XY. Condition on the pair being primitive modulo ℓ; its probability is $1 - \ell^{-2}$. For $1 \le t \le e$ the event $v_\ell(F_M) \ge t$ consists of the two disjoint cases in which one coordinate is a unit and the other is divisible by ℓ^t. Hence

$$\Pr(v_\ell(F_M) \ge t \mid \text{primitive}) = \frac{2\ell^{-t}(1-\ell^{-1})}{1-\ell^{-2}} = \frac{2}{(\ell+1)\ell^{t-1}}.$$

Taking successive differences gives precisely the $\ell \equiv 1$ distribution in the table. At 2, when both diagonal entries are even, the preceding theorem gives a forced 2 and $e = 1$. Otherwise $F_M(x,y) \pmod 2 = Ax + Dy$ is a nonzero linear form. Of the three primitive pairs modulo 2, exactly one makes it zero. By the same theorem $4 \nmid \gcd(F_M, k)$, giving the two probabilities. The Chinese remainder theorem makes these local distributions independent when a primitive residue pair is chosen uniformly modulo k, and proves (4).

For completeness, each primitive residue class $(a,b) \pmod k$ occurs with the same asymptotic frequency among $\mathcal{V}_X$. Möbius inversion in the common divisor, using $\gcd(a,b,k) = 1$, gives

$$\#\{(x,y) \in \mathcal{V}_X : (x,y) \equiv (a,b) \pmod k\} = \frac{X^2}{k^2\zeta(2)} \prod_{\ell \mid k} (1-\ell^{-2})^{-1} + O_k(X \log(2X)).$$

Indeed only divisors coprime to k survive; for each such divisor d, each coordinate has $X/(kd) + O(1)$ choices, and summing the resulting errors costs $O_k(X\log(2X))$. There are $k^2 \prod_{\ell \mid k}(1-\ell^{-2})$ primitive residue classes modulo k. Summing over those with the specified cancellation factor yields (5). The usual $\#\mathcal{V}_X = X^2/\zeta(2) + O(X\log(2X))$ finishes the relative frequency statement. □

Remark 4.2 (Two exact spectra and probability laws)**.** *For* $M = \left(\begin{smallmatrix} 1 & 10 \\ 0 & 1 \end{smallmatrix}\right)$*, we have* $k = 10$ *and*

$$\rho(1) = 4/9, \qquad \rho(2) = 2/9, \qquad \rho(5) = 2/9, \qquad \rho(10) = 1/9.$$

For $M = \left(\begin{smallmatrix} 1 & 25 \\ 0 & 1 \end{smallmatrix}\right)$*,* $\rho(1) = 2/3$*,* $\rho(5) = 4/15$ *and* $\rho(25) = 1/15$*. The exception* $M = \left(\begin{smallmatrix} 2 & 1 \\ 3 & 2 \end{smallmatrix}\right)$ *has* $k = 2$ *and* $\rho(2) = 1$*.*

Remark 4.3 (Relation to classical density methods)**.** *The split/anisotropic local classification and Möbius inversion are standard ingredients of arithmetic theory. The result here packages them as the* full distribution of exact denominator cancellations *for the canonical unimodular interpolation problem, not as a claim to introduce new local-density tools. It is different from counting distinct integers represented by a quadratic form; compare general local-density frameworks such as [6].*

5 A fixed factor for a family of adjacent-swap defects

Let $G, H \in \mathrm{Sym}_2(\mathbb{Q})$ with $\det G = \det H = 1$. Set

$$A = GH, \quad A^T = HG, \quad S = \tfrac{1}{2}(A + A^T), \quad K = \tfrac{1}{2}(A - A^T), \quad d = A_{12} - A_{21}.$$

For symmetric P define

$$E_P(v) = Q_P(Av) - Q_P(A^T v).$$

Direct polarization gives

$$E_P(v) = 4(Sv)^T P(Kv), \qquad K = -\frac{d}{2}J, \qquad \det S = 1 - \frac{d^2}{4}. \tag{6}$$

Proposition 5.1 (Fricke commutator interpretation)**.** *For the symmetric determinant-one pair G, H above, let $[G, H] = GHG^{-1}H^{-1}$. Then*

$$\boxed{\operatorname{tr}[G, H] = 2 - d^2.} \tag{7}$$

In particular $|d| = 2$ if and only if $\operatorname{tr}[G, H] = -2$. Consequently, if $x = \operatorname{tr} G$, $y = \operatorname{tr} H$, and $z = \operatorname{tr}(GH)$, this case satisfies the classical Fricke–Markoff equation

$$x^2 + y^2 + z^2 = xyz. \tag{8}$$

Proof. Put $U = GH$ and $V = HG$, so $\det U = \det V = 1$ and $U - V = -dJ$. As $\det(U - V) = \det(I - U^{-1}V)$ and $\det(U^{-1}V) = 1$, the 2×2 determinant identity gives

$$d^2 = 2 - \operatorname{tr}(U^{-1}V) = 2 - \operatorname{tr}(GHG^{-1}H^{-1}).$$

The last trace equality follows from invariance under conjugation and inversion for determinant-one 2×2 matrices. The remaining assertion follows from the classical Fricke trace formula $\operatorname{tr}[G, H] = x^2 + y^2 + z^2 - xyz - 2$; see, for instance, [3, 4]. □

Theorem 5.2 (Universal linear-factor criterion)**.** *Assume $GH \neq HG$. A nonzero rational homogeneous linear form $L(v)$ divides $E_P(v)$ for* every *rational symmetric P if and only if $|d| = 2$, equivalently $\operatorname{tr}[G, H] = -2$. When this occurs $S = \lambda zz^T$ has rank one for a rational nonzero vector z and $\lambda \in \mathbb{Q}^\times$, and*

$$E_P(v) = 4\lambda(z^T v)(z^T PKv)$$

for all symmetric P.

Proof. Noncommutativity implies $d \neq 0$, since $A - A^T$ is a 2×2 skew matrix. If $d^2 = 4$ then $\det S = 0$. Moreover $S \neq 0$: real symmetric determinant-one matrices are either positive or negative definite, so GH has nonzero real eigenvalues of the same sign, while a nonzero skew matrix of determinant one has purely imaginary eigenvalues. Thus S has rank one, giving the factorization by (6).

Conversely, let L divide all E_P, and let $v \in \mathbb{Q}^2$ be a nonzero vector in $L(v) = 0$. Varying P over all symmetric matrices gives

$$(Av)(Av)^T = (A^T v)(A^T v)^T.$$

Consequently $Av = \pm A^T v$. The plus sign is impossible, since $(A - A^T) = -dJ$ is invertible. Therefore $(A + A^T)v = 0$. It follows that $\det S = 0$, and (6) yields $d^2 = 4$. □

Remark 5.3. *The condition $\operatorname{tr}[G, H] = -2$ is the classical parabolic commutator locus of Fricke–Markoff theory, not a newly discovered trace locus. What is shown here is its equivalence with the particular* quadratic-observable common-factor property*.*

Remark 5.4. *This statement concerns a P-independent factor shared by an entire three-dimensional linear family of quadratic forms. It does not assert that an individual E_P factors over $\mathbb{Q}$ only when $|d| = 2$; individual factorization is governed by the usual binary quadratic discriminant.*

6 A structural return theorem for free positive monoids

The common factor in Theorem 5.2 need not be regarded merely as a polynomial curiosity. It singles out a natural positive state vector and hence a family of canonical return observables.

Theorem 6.1 (Rank-one swap criterion implies programmable word returns)**.** *Let $G, H \in \mathrm{Sym}_2(\mathbb{Z})$ have determinant one and strictly positive entries. Assume that G and H do not commute. Put*

$$A = GH, \quad d = A_{12} - A_{21}, \quad S = (A + A^T)/2.$$

Suppose $|d| = 2$. Then $S = \lambda u u^T$ for some $\lambda \in \mathbb{Q}_{>0}$ and some strictly positive primitive vector $u \in \mathbb{Z}^2$. For a word $w \in \{0,1\}^*$, let $M(w)$ be its matrix product (with $0 \mapsto G$, $1 \mapsto H$) and put

$$q(w) = M(w)^T u, \qquad f(w) = u^T q(w), \qquad P_w = P(M(w), u).$$

If the map $w \mapsto q(w)$ is injective, then for all words s, w,

$$Q_{P_w}(GHq(s)) - Q_{P_w}(HGq(s)) = -2\lambda d\, f(s)\, q(w)^T J q(s). \tag{9}$$

In particular the quadratic swap defect vanishes precisely when $s = w$, and

$$P_w \in \mathrm{Sym}_2(\mathbb{Z}) \quad \Longleftrightarrow \quad w = w^R.$$

Proof. Theorem 5.2 gives $\det S = 0$ and $S \neq 0$. Because G, H have strictly positive entries, S has strictly positive entries and is thus a positive semidefinite rank-one rational matrix. Its one-dimensional image is generated by a strictly positive primitive integer vector u and $S = \lambda u u^T$ with $\lambda > 0$ rational.

The skew part is $K = -dJ/2$. Polarization (6) and $P_w u = q(w)$ show

$$E_{P_w}(q(s)) = 4\lambda (u^T q(s))\, u^T P_w(-dJ/2) q(s) = -2\lambda d f(s) q(w)^T J q(s).$$

All the word states are positive primitive vectors, since u is primitive and $M(w)$ is integral unimodular. Vanishing of the alternating product is therefore equivalent to $q(s) = q(w)$; injectivity gives $s = w$. Finally the matrix word map is also injective (equality of matrices would imply equality of states). Corollary 2.4 applies and gives the integrality–palindrome equivalence. □

Remark 6.2. *The hypotheses are substantive. The relation $|d| = 2$ controls a universal factorization, whereas positivity and injectivity turn its determinant zero locus into a single* word *rather than an entire projective rational line. Without an injective state decoder, different words could produce the same primitive state; without the rank-one swap condition, even a fixed common linear factor need not exist.*

Remark 6.3 (The balanced pair is a classical Markoff-trace point)**.** *For the pair in* (10)*, the trace triple is* $(3,3,6)$*, and $3^2 + 3^2 + 6^2 = 3 \cdot 3 \cdot 6$. The original Cohn pair with $X = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ and $Y = \left(\begin{smallmatrix} 5 & 2 \\ 2 & 1 \end{smallmatrix}\right)$ has trace triple* $(3,6,15)$*, which satisfies the same Fricke–Markoff cubic. The two examples are therefore not evidence for a new trace class.*

7 Prescribing a unique return word

We exhibit a concrete pair distinct from the classical Cohn generators:

$$G = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \qquad H = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}, \qquad u = \begin{pmatrix} 1 \\ 1 \end{pmatrix}. \tag{10}$$

For a word $w \in \{0,1\}^*$ let $M(\varepsilon) = I$, $M(w0) = M(w)G$, and $M(w1) = M(w)H$. Define the positive primitive column $q(w) = M(w)^T u$ and $f(w) = u^T q(w)$. Let $P_w = P(M(w), u)$ be the canonical form of (1).

Lemma 7.1 (Exact state decoder)**.** *The map $w \mapsto q(w)$ is injective. A nonempty word ends in* 0 *precisely when $q_1(w) > q_2(w)$, and ends in* 1 *precisely when $q_1(w) < q_2(w)$.*

Proof. If $q = (x, y)^T$ has positive coordinates, then $Gq = (2x + y, x + y)^T$ has first coordinate larger by x, while $Hq = (x + y, x + 2y)^T$ has second coordinate larger by y. Starting at $(1, 1)$, the last letter is uniquely determined by the sign of the coordinate difference. Invert using G^{-1} or H^{-1}; on a genuine state this recovers the positive preceding state and strictly decreases the coordinate sum. Iteration reaches the unique initial state $(1, 1)$, hence uniquely recovers the word. □

Theorem 7.2 (Programmable single-word swap returns)**.** *For the generators* (10)*, every word w has a positive-definite determinant-one rational form P_w with*

$$Q_{P_w}(GHq(s)) - Q_{P_w}(HGq(s)) = -12f(s)\, q(w)^T Jq(s) \tag{11}$$

for every word s. In particular,

$$Q_{P_w}(GHq(s)) = Q_{P_w}(HGq(s)) \quad \Longleftrightarrow \quad s = w.$$

Furthermore,

$$P_w \text{ is integral} \quad \Longleftrightarrow \quad w = w^R.$$

Proof. Here

$$GH = \begin{pmatrix} 3 & 4 \\ 2 & 3 \end{pmatrix}, \qquad HG = \begin{pmatrix} 3 & 2 \\ 4 & 3 \end{pmatrix}, \qquad S = 3uu^T, \quad K = -J.$$

Equation (6) gives $E_P(v) = -12(u^T v)(u^T PJv)$. Substituting $P = P_w$ and using $P_w u = q(w)$ yields (11). Because $f(s) > 0$, a zero is equivalent to $q(w)^T Jq(s) = 0$. Both vectors are positive primitive integral vectors, since they lie in an $SL_2(\mathbb{Z})$ orbit of the primitive seed. Positive collinearity of such vectors forces equality. Lemma 7.1 then gives $s = w$, and the converse is immediate. The integrality criterion follows from Corollary 2.4. □

For example $w = 01$ has $q(w) = (5, 7)^T$ and $f(w) = 12$; consequently

$$P_{01} = \frac{1}{6}\begin{pmatrix} 13 & 17 \\ 17 & 25 \end{pmatrix}, \qquad \operatorname{den}(P_{01}) = 6,$$

with 01 as its only swap-zero word. For the palindromic one-letter words, $P_0 = G$ and $P_1 = H$.

8 Scope of the conclusions

Theorem 2.1 does not require a word monoid, nonnegative entries of M, or a Markoff condition. Corollary 2.2 additionally uses positivity of both the matrix entries and the seed; Corollary 2.4 uses injectivity of the matrix word representation. Theorem 5.2 classifies a *common* factor for all quadratic observables attached to two symmetric determinant-one generators. Theorem 7.2 is a genuine *orbit* statement for the explicitly decoded generators (10), rather than an identity on the entire plane or a global uniqueness claim for Markoff numbers.

The Fricke trace locus, Markoff cubic, and Cohn–Christoffel palindrome descriptions are classical [1, 5, 2, 3, 4]. The distinct arithmetical content in this arrangement is the exact denominator/reversal formula and prescribed-return realization across well-specified matrix classes. These formulas have short elementary proofs and may follow from more general established form-theoretic results; no assertion of novelty priority is made solely on the basis of the present comparison.

9 Finite verification and its scope

Exact-arithmetic consistency checks evaluated the least-denominator formula on 27,664 positive primitive seed/matrix instances drawn from 304 nonsymmetric nonnegative unimodular matrices with entries between 0 and 16. For all 304 matrices, direct counting of primitive residue classes modulo $|B - C|$ reproduced the predicted local cancellation distributions. In the explicit two-generator word model, all 511 words of length at most 8 had distinct states; the quadratic swap identity and its unique-zero criterion agreed on all 261,121 ordered word pairs, and the integrality–palindrome equivalence held for the same 511 words. No discrepancy was found in these bounded checks. These finite calculations are consistency tests, not substitutes for the symbolic proofs or a certification of the general statements outside the tested ranges. No computational files are required for the proofs in this article.

Statements and Declarations

Author contributions. Alp Eren Bütün: conceptualization, methodology, formal analysis, investigation, software, validation, writing—original draft, and writing—review and editing.
Funding. No external funding was received for this research.
Competing interests. The author declares no relevant financial or non-financial competing interests.
Data availability. This is a mathematical study and uses no empirical or personal dataset; the relevant definitions, statements, constructions and symbolic proofs are contained in the article.
Ethics approval and consent to participate. Not applicable.
Consent for publication. Not applicable.

References

[1] H. Cohn, *Markoff Forms and Primitive Words*, Math. Ann. **196** (1972), 8–22. `https://eudml.org/doc/162238`.

[2] C. Reutenauer, *On quadratic numbers and forms, and Markoff theory*, J. Number Theory **227** (2021), 265–305. doi:10.1016/j.jnt.2021.03.005.

[3] G. McShane and H. Parlier, *Multiplicities of simple closed geodesics and hypersurfaces in Teichmüller space*, Geom. Topol. **12** (2008), 1883–1919. doi:10.2140/gt.2008.12.1883.

[4] Z. Greenberg, D. Kaufman and A. Wienhard, *SL_2-like properties of matrices over noncommutative rings and generalizations of Markov numbers*, Adv. Math. **482** (2025), 110556. doi:10.1016/j.aim.2025.110556.

[5] C. Reutenauer, *From Christoffel Words to Markoff Numbers*, 2nd ed., Oxford University Press, 2026. doi:10.1093/9780197907535.001.0001.

[6] P. L. Clark, P. Pollack, J. Rouse and K. Thompson, *Densities of integer sets represented by quadratic forms*, J. Number Theory **256** (2024), 290–328. doi:10.1016/j.jnt.2023.09.010.